FEMM Current Flow File Formats

Input (`.fec`) and Output (`.anc`) Specifications

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September 13, 2026

Contents

1 Overview

The FEMM current flow solver (`csolv`) reads a problem definition from a `.fec` file and writes the solution to a `.anc` file. The `.anc` file begins with a verbatim copy of the input `.fec` file, followed by a `[Solution]` section containing the finite element mesh and computed complex voltage values.

Current flow is an AC problem. The solver computes the complex voltage phasor $\hat{V}$ at each mesh node, from which the electric field $E = -\nabla V$ and current density $J = \sigma E + j\omega\varepsilon E$ are derived. All solution quantities (voltages, currents, fields) are complex-valued.

All floating-point values are written with 17 significant digits (`%.17g` format). Fields within a line are separated by tabs or spaces. Section headers use square brackets (`[Tag]`); property block delimiters use angle brackets (`<BeginBlock>` / `<EndBlock>`). String comparisons during parsing are case-insensitive.

2 The `.fec` Input File

2.1 Global Problem Parameters

Each parameter occupies one line in the format `[Tag] = value`. All are optional; defaults are applied if missing.

Tag	Type	Default	Description
`[Format]`	int	1	File format version
`[Frequency]`	double	60	AC frequency (Hz)
`[Precision]`	double	1e-8	Solver convergence tolerance
`[MinAngle]`	double	30	Minimum mesh element angle (degrees)
`[DoSmartMesh]`	int	1	Adaptive meshing (0=off, 1=on)
`[Depth]`	double	1.0	Extrusion depth (planar problems)
`[LengthUnits]`	string	millimeters	inches, millimeters, centimeters, meters, mils, microns
`[ProblemType]`	string	planar	planar or axisymmetric
`[Coordinates]`	string	cartesian	cartesian or polar
`[Comment]`	string	(empty)	Problem description (quoted; `\n` = newline)
`[ACSolver]`	int	0	Always 0 (compatibility field)

The `[Frequency]` field distinguishes current flow from electrostatics. At $f = 0$, the displacement current term vanishes and the problem reduces to pure DC conduction. The solver internally converts frequency to angular frequency: $\omega = 2\pi f$.

2.1.1 Axisymmetric External Region

Written only when `ProblemType` is axisymmetric and both `extRo` and `extRi` are nonzero:

Tag	Type	Description
`[extZo]`	double	Outer Z coordinate

[extRo]	double	Outer radius
[extRi]	double	Inner radius

2.2 Point (Nodal) Properties

All point property values are complex (separate real and imaginary parts) to support AC excitation.

```
[PointProps] = <count>
<BeginPoint>
  <PointName> = "name"
  <vpr> = 0      prescribed voltage, real (V)
  <vpi> = 0      prescribed voltage, imag (V)
  <qpr> = 0      point current source, real (A)
  <qpi> = 0      point current source, imag (A)
<EndPoint>
... repeated count times
```

2.3 Boundary Properties

All boundary values are complex-valued.

```
[BdryProps] = <count>
<BeginBdry>
  <BdryName> = "name"
  <BdryType> = 0
  <vsr> = 0        fixed voltage, real (V)
  <vsi> = 0        fixed voltage, imag (V)
  <qsr> = 0        surface current density, real (A/m)
  <qsi> = 0        surface current density, imag (A/m)
  <c0r> = 0        mixed BC coefficient, real
  <c0i> = 0        mixed BC coefficient, imag
  <c1r> = 0        mixed BC coefficient, real
  <c1i> = 0        mixed BC coefficient, imag
<EndBdry>
```

Boundary types:

- 0 = fixed voltage (Dirichlet): $V = V_s$
- 1 = mixed (Robin): $\hat{n} \cdot J = c_0 + c_1 V$
- 2 = surface current density (Neumann)
- 3 = periodic
- 4 = antiperiodic

2.4 Material (Block) Properties

Current flow materials combine electrical conductivity and dielectric properties. The solver constructs an effective complex admittivity from these parameters:

$$K_x = \frac{\sigma_x}{\varepsilon_0} + j\omega\,\varepsilon_{r,x}\,e^{-j\arctan(\tan\delta_x)}$$

and similarly for K_y.

```
[BlockProps] = <count>
<BeginBlock>
  <BlockName> = "name"
  <ox> = 0        DC conductivity, x-direction (MS/m)
  <oy> = 0        DC conductivity, y-direction (MS/m)
  <ex> = 1        relative permittivity, x-direction
  <ey> = 1        relative permittivity, y-direction
  <ltx> = 0       loss tangent, x-direction (tan delta)
  <lty> = 0       loss tangent, y-direction (tan delta)
<EndBlock>
```

- Conductivity σ is in megasiemens per meter (MS/m).
- Loss tangent $\tan\delta$ creates a complex permittivity that models dielectric loss. Zero means lossless dielectric.
- At $f = 0$ (DC), the displacement current term vanishes and only the conductivity term remains.
- At $\sigma = 0$ (perfect insulator), only displacement current flows.

2.5 Conductor Properties

Conductors enforce equipotential constraints or prescribe total current on electrode surfaces. All values are complex.

```
[ConductorProps] = <count>
<BeginConductor>
  <ConductorName> = "name"
  <vcr> = 0         prescribed voltage, real (V)
  <vci> = 0         prescribed voltage, imag (V)
  <qcr> = 0         prescribed total current, real (A)
  <qci> = 0         prescribed total current, imag (A)
  <ConductorType> = 0
<EndConductor>
```

Conductor types:

- 0 = prescribed total current ($\hat{q}$ given; solver computes $\hat{V}$)
- 1 = prescribed voltage ($\hat{V}$ given; solver computes $\hat{q}$)

2.6 Geometry

2.6.1 Nodes

```
[NumPoints] = <count>
x   y   propIdx   inGroup   conductorIdx   [lfs]
```

- `x`, `y`: coordinates (double)
- `propIdx`: index into point property list (1-based; 0 = none)
- `inGroup`: group number (int)
- `conductorIdx`: index into conductor list (1-based; 0 = none)
- `lfs`: local feature size constraint (double, optional; -1 = auto)

2.6.2 Line Segments

```
[NumSegments] = <count>
n0   n1   maxSideLength   propIdx   hidden   inGroup   conductorIdx
```

- `n0`, `n1`: endpoint node indices (0-based)
- `maxSideLength`: mesh size constraint (double; -1 = auto)
- `propIdx`: boundary property index (1-based; 0 = none)
- `hidden`: 0 = visible, 1 = suppressed from mesh
- `inGroup`: group number
- `conductorIdx`: conductor index (1-based; 0 = none)

2.6.3 Arc Segments

```
[NumArcSegments] = <count>
n0  n1  arcAngle maxSideLength propIdx hidden inGroup conductorIdx mySideLength
```

- `n0`, `n1`: endpoint node indices (0-based)
- `arcAngle`: arc span (degrees)
- `maxSideLength`: mesh constraint (double)
- `propIdx`: boundary property index (1-based; 0 = none)
- `hidden`: 0 = visible, 1 = suppressed
- `inGroup`: group number
- `conductorIdx`: conductor index (1-based; 0 = none)
- `mySideLength`: computed side length (0 = auto)

2.6.4 Holes

```
[NumHoles] = <count>
x   y   inGroup
```

2.6.5 Block Labels

```
[NumBlockLabels] = <count>
x   y   blockIdx   meshSize   inGroup   extFlags
```

- `x`, `y`: label position (double)
- `blockIdx`: material index (1-based; 0 = unassigned)
- `meshSize`: element size (double; ≤ 0 = auto). Stored as diameter; converted to area: $A = \pi d^2/4$
- `inGroup`: group number
- `extFlags`: bit 0 = external region, bit 1 = default region

2.7 Differences from Other Physics Formats

- `[Frequency]` is always present and typically nonzero (default 60 Hz), unlike electrostatics (always 0) and heat flow (no frequency).
- All point properties, boundary values, and conductor properties have separate real/imaginary parts (complex AC phasors).

- Material properties include both conductivity (σ_x, σ_y) and permittivity (ε_x, ε_y) with loss tangent — unique to current flow.
- No nonlinear material curves (no B-H or T-K equivalent).
- No transient fields (`[dT]`, `[PrevSoln]` are heat-flow-specific).

3 The `.anc` Output File

The `.anc` file begins with a verbatim copy of the input `.fec` file (all sections described in Section 2), followed by the solver's solution data.

3.1 Solution Header

Immediately after the echoed `.fec` content:

```
[Solution]
```

3.2 Mesh Nodes

```
<NumNodes>
```

followed by `NumNodes` lines:

```
x    y    V_re    V_im    Q
```

- `x`, `y`: node coordinates in model units
- `V_re`: real part of complex voltage phasor (V)
- `V_im`: imaginary part of complex voltage phasor (V)
- `Q`: boundary/conductor marker (int):
  - -2 = interior node (no boundary condition)
  - -1 = fixed voltage boundary
  - $0 \ldots N-1$ = conductor index

The voltage phasor represents the steady-state AC solution. The instantaneous voltage at any point is $V(t) = |\hat{V}| \cos(\omega t + \arg \hat{V})$.

Node indices are 0-based throughout the solution section.

3.3 Mesh Elements

```
<NumElements>
```

followed by `NumElements` lines:

```
p0    p1    p2    labelIdx
```

- `p0`, `p1`, `p2`: vertex node indices (0-based)
- `labelIdx`: block label index (0-based, into the block label list from the `.fec` header)

3.4 Conductor Solution Data

```
<NumConductors>
```

followed by `NumConductors` lines:

```
V_re   V_im   q_re   q_im
```

- `V_re`, `V_im`: complex voltage on the conductor (V). For prescribed-voltage conductors, this is the prescribed value; for prescribed-current conductors, this is the computed value.
- `q_re`, `q_im`: complex total current through the conductor (A). For prescribed-current conductors, this is the prescribed value; for prescribed-voltage conductors, this is the computed value.

4 Index Conventions

- **Property indices** (materials, boundaries, conductors, point properties) are **1-based** in the file. An index of 0 means "no property assigned." The parser converts to 0-based internally.
- **Node indices** (in segments, arcs, and the solution section) are **0-based**.
- **Block label indices** in the solution element list are **0-based**, referencing the block label array from the `.fec` header.

5 Unit Conversion

The `[LengthUnits]` tag determines the coordinate system. All coordinates in the file are in the specified units. Conversion to meters:

Unit	Meters per unit
inches	0.0254
millimeters	0.001
centimeters	0.01
meters	1.0
mils	2.54×10^{-5}
microns	10^{-6}